Bayesian Estimation of Nonlinear DSGE Models with Neural-Network Surrogate Solvers

A Stochastic Extended Path to HMC pipeline

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Views expressed are my own and do not necessarily represent the IMF.

Main Takeaway First

Do not put the global nonlinear solver inside the posterior sampler.

What I do

I solve the nonlinear model offline with Stochastic Extended Path, train a differentiable residual surrogate, and use that surrogate inside HMC.

Why it matters

This turns nonlinear solution cost from a per-likelihood cost into an offline training and validation cost.

The contribution is computational: make Bayesian nonlinear DSGE estimation operational without pretending the nonlinear solver is cheap.

The Hook

Nonlinear DSGE models are economically natural and computationally awkward.

Economists care about nonlinear states.

- Effective lower bounds.
- Occasionally binding constraints.
- Large shocks and crisis periods.
- Curvature in pricing, wages, investment, and utilization.

But estimation usually retreats to linear tools.

- Perturbation is fast.
- Kalman filtering is stable.
- The full nonlinear likelihood is too expensive.
- Particle filters can be fragile in high dimension.

The question is not whether nonlinearities matter. The question is whether we can estimate them at scale.

Question

How can we estimate nonlinear DSGE models without solving the model at every posterior draw?

The empirical problem

Medium-scale DSGE models are useful because they connect mechanisms, shocks, and observables. The same structure makes nonlinear estimation expensive.

What we want

- Structural parameters with posterior uncertainty.
- Likelihood comparisons across nonlinear mechanisms.
- Diagnostics that say when linearization is misleading.

What blocks us

- Repeated nonlinear solves.
- Numerical failures away from the mode.
- Slow or noisy likelihood evaluations.

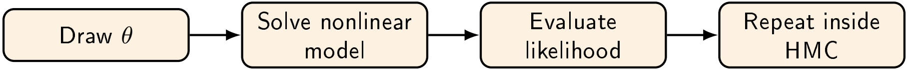
Literature Review

- **Nonlinear DSGE likelihoods and filters:**
  - Fernandez-Villaverde and Rubio-Ramirez (2007); Fernandez-Villaverde, Rubio-Ramirez, and Schorfheide (2016); Herbst and Schorfheide (2019).
  - These papers make nonlinear likelihood-based estimation credible, but repeated nonlinear filtering remains costly in medium-scale models.
- **Occasionally binding constraints and piecewise-linear filtering:**
  - Guerrieri and Iacoviello (2015); Cuba-Borda, Guerrieri, Iacoviello, and Zhong (2019); Aruoba, Cuba-Borda, Higa-Flores, Schorfheide, and Villalvazo (2021).
  - These methods are stable and useful for OBCs, but keep the solution close to linear or piecewise-linear objects.
- **Neural network solvers and estimation:**
  - Maliar, Maliar, and Winant (2021); Azinovic, Gaegauf, and Scheidegger (2022); Kase, Melosi, and Rottner (2025, MKR); Naubert (2025).
  - Closest paper: MKR use neural networks to make nonlinear structural estimation feasible in heterogeneous-agent models.
- **This paper: improvements relative to MKR and related work**
  - **SEP teacher:** the NN is trained and audited against direct global nonlinear solves.
  - **Residual target:** learn $g^{\mathrm{SEP}} - g^{\mathrm{ROM1}}$, so the method collapses to ROM1 when linearization is enough.
  - **Gated likelihood:** the correction is active only where support has been checked.
  - **Economic output:** the residual identifies which DSGE mechanisms create the nonlinear gap.

MKR denotes Kase, Melosi, and Rottner, "Estimating Nonlinear Heterogeneous Agent Models with Neural Networks" (2025).

The Computational Problem

The nonlinear solver sits in the wrong loop.



Repeated many times inside HMC.

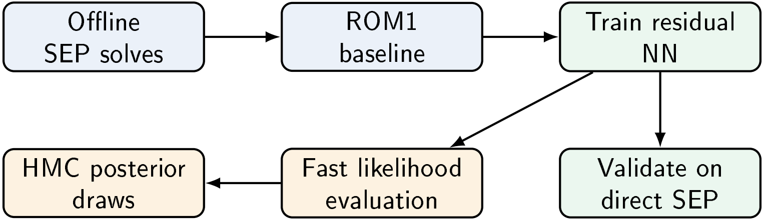
The nested-loop bottleneck

$$\theta \longmapsto g_\theta^{\mathrm{SEP}} \longmapsto p(y_{1:T} \mid \theta) \longmapsto \text{MCMC proposal}$$

The expensive object is recomputed exactly where the sampler needs speed.

The Solution in One Picture

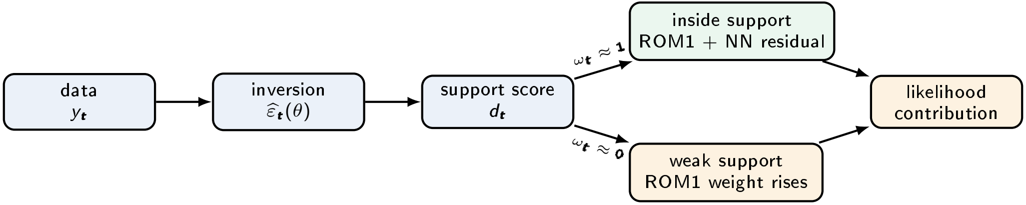
Move the expensive nonlinear solve offline.



SEP is the teacher. ROM1 is the baseline. The NN learns the nonlinear correction.

How the Gate Operates

The gate is an audit rule inside the likelihood.



Soft transition map

$$\widehat{g}_t = g_\theta^{\mathrm{ROM1}} + \omega_t r_\phi, \qquad 0 \le \omega_t \le 1.$$

What the gate reports

- Where the NN correction is active.
- Where the linear fallback dominates.
- Which periods need direct SEP audit.

Roadmap

1. Takeaway, motivation, and question ✓
2. The computational bottleneck ✓
3. The SEP–ROM1–NN–HMC pipeline
4. Small-model validation with a hard nonlinearity
5. Smets–Wouters extension and open work

Timing guide: 6 min setup, 12 min pipeline, 7 min validation, 4 min medium-scale extension, 1 min close.

Step 0: Choose the Estimation Target

Be explicit about the object before approximating it.

Start with a nonlinear model, a parameter vector θ, observed data $y_{1:T}$, and a likelihood or profiled likelihood criterion.

Inputs

- Model equations.
- Parameter support.
- Shock process and observables.
- Measurement-error or inversion design.

Audit trail

- Which parameters vary?
- What region is trained?
- What likelihood is approximated?
- What is held fixed?

Step 1: Solve the Nonlinear Model Offline

Use SEP as an offline teacher.

For sampled triples (s, ε, θ), compute the nonlinear transition

$$s^{+}_{\mathrm{SEP}} = g^{\mathrm{SEP}}_{\theta}(s, \varepsilon).$$

Why SEP?

- Keeps the nonlinear equations.
- Handles occasionally binding constraints.
- Gives direct objects for validation.

What must be reported

- Solver tolerances.
- Failed and successful points.
- Shock and parameter support.
- Initialization strategy.

Step 2: Compute the Linear Baseline at the Same Points

Use the linear solution as the baseline, not as the answer.

At the same inputs, compute

$$s^{+}_{\mathrm{ROM1}} = g_{\theta}^{\mathrm{ROM1}}(s, \varepsilon).$$

Reason

The linear solution is cheap and often accurate near the steady state. The NN should not waste capacity relearning what ROM1 already does well.

$$\text{Nonlinear object} = \text{linear baseline} + \text{nonlinear correction}.$$

Step 3: Train the Residual Surrogate

Train the nonlinear correction.

Train r_ϕ on the residual

$$r_\phi(s,\varepsilon,\theta) \approx g_\theta^{\mathrm{SEP}}(s,\varepsilon) - g_\theta^{\mathrm{ROM1}}(s,\varepsilon).$$

Why residual learning

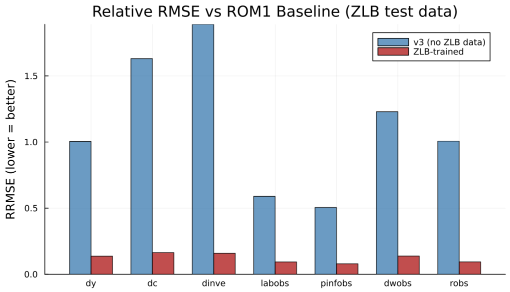
- Smaller target.
- Easier extrapolation around the linear region.
- Directly measures what linearization misses.

Resulting transition

$$g_\phi(s,\varepsilon,\theta) = g_\theta^{\mathrm{ROM1}}(s,\varepsilon) + r_\phi(s,\varepsilon,\theta).$$

Step 4: Validate the Surrogate Before Estimation

Do not estimate before the surrogate passes basic audits.



Checks

- Held-out direct SEP points.
- RMSE and relative RMSE by observable.
- Regime-specific accuracy: binding, non-binding, near-boundary.
- Finite likelihood and finite-gradient checks.

Rule

The surrogate is only credible on the support where it has been audited.

Step 5: Build the Likelihood

The likelihood determines what the validation has to prove.

Small validation model

Use a direct measurement-error likelihood. Compare direct SEP-HMC with surrogate-HMC on the same synthetic data.

Medium-scale empirical model

Use an inversion step to recover shocks period by period, then propagate with

$$\widehat{s}_{t+1} = g_{\theta}^{\mathrm{ROM1}}(s_t, \widehat{\varepsilon}_{t+1}) + r_{\phi}(s_t, \widehat{\varepsilon}_{t+1}, \theta).$$

The medium-scale target is a profiled hybrid criterion. That caveat should be explicit.

Step 6: Sample Parameters with HMC

The differentiable surrogate is what makes HMC practical.

Why HMC

- Uses gradients to move through correlated DSGE posteriors.
- Needs many fast target evaluations.
- Works naturally with differentiable surrogates.

What is matched in validation

- Same data.
- Same priors.
- Same transforms.
- Same seeds and HMC settings.

Comparison

The surrogate posterior should match the direct SEP posterior within Monte Carlo error, with no systematic directional distortion.

Step 7: Audit the Whole Pipeline

A surrogate paper lives or dies by its audit trail.

Object	Question	Failure mode
SEP support	Did the solver cover the region?	Hidden selection
Surrogate fit	Is the map accurate where used?	Biased likelihood
Gradients	Are target derivatives finite?	Bad HMC geometry
HMC chains	Do chains mix and agree?	Unreliable posterior
Direct audits	Does SEP confirm key cells?	Unsupported extrapolation

The replication package should make these checks mechanical, not narrative.

The Complete Pipeline

The method is a sequence of auditable approximations.

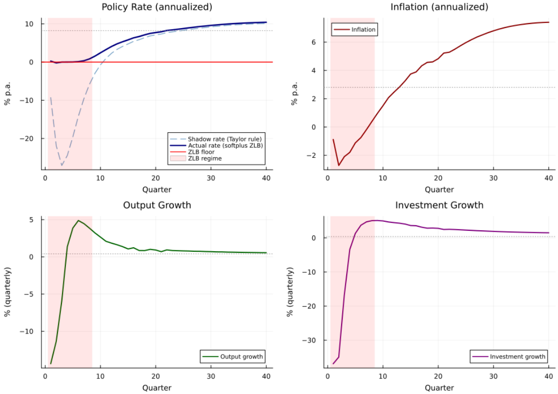
1. Define model, observables, parameter support, and likelihood target.
2. Draw training states, shocks, and parameters.
3. Solve direct nonlinear transitions with SEP.
4. Compute ROM1 transitions at identical inputs.
5. Train the NN on the SEP–ROM1 residual.
6. Validate the map on held-out direct SEP points.
7. Insert $g^{\mathrm{ROM1}} + r_\phi$ into the likelihood.
8. Run HMC and audit the posterior against direct SEP where feasible.

The discipline

Every empirical number should trace back to raw artifacts: support, errors, diagnostics, and sampler output.

Validation: A Hard Nonlinearity

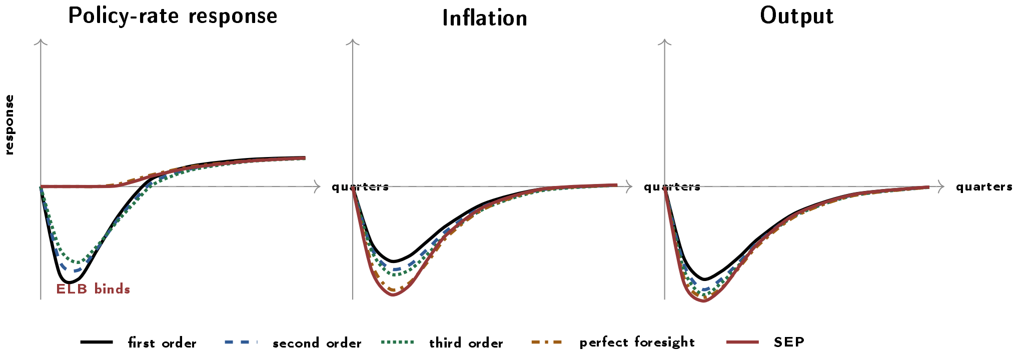
The validation needs a region where linearization actually fails.



Same shocks, nonlinear occasionally-binding model versus linearized model.

IRF Ladder: Local to Global

For a real results slide, run one adverse demand shock through every solver on the same calibration.



Schematic template for an adverse demand/natural-rate shock in which the shadow policy rate falls and the ELB binds. This is not a policy-rate shock result. For evidence, the same shock vector and calibration must be run through each solver and the artifact path reported.

Validation: What the Small Model Shows

The small model is a pipeline test, not a victory lap.

Design

- Nonlinear New Keynesian prototype.
- Synthetic data generated by the nonlinear model.
- Selected shock-scale parameters estimated.
- Direct SEP-HMC compared to surrogate-HMC.

Interpretation

- The pipeline can recover parameters in a controlled nonlinear setting.
- The validation is local to the model and support.
- It is a feasibility proof, not a universal theorem.

Scaling Up: Smets–Wouters

The medium-scale model is where the computational problem becomes real.

Medium-scale target

Smets–Wouters style model with investment adjustment costs, variable utilization, Kimball pricing and wages, and occasionally binding constraints.

Why this is useful

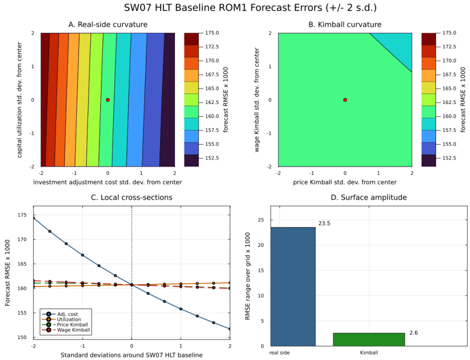
- Standard empirical DSGE laboratory.
- Rich enough to stress nonlinear mechanisms.
- Economically interpretable posterior shifts.

Why this is hard

- Higher-dimensional state.
- Fragile nonlinear solves.
- Strong need for out-of-support diagnostics.

Where the Nonlinear Forecast Error Comes From

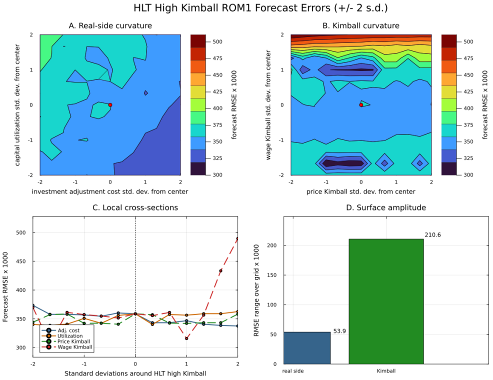
Around the posterior region, real-side curvature is the first suspect.



Around the baseline/posterior region, investment and utilization curvature move the SEP–ROM1 forecast error more than Kimball curvature.

Stress Test: High-Kimball Region

Kimball can dominate, but only in the right neighborhood.



What changes

If the calibration is deliberately moved to a high price/wage-curvature neighborhood, Kimball curvature can dominate the local forecast-error surface.

What this means

The investment-channel result is a posterior-region finding, not a global claim about every DSGE calibration.

Computational Accounting

The computational gain is amortization.

Task	Timing	Role
SEP solves	offline	expensive teacher
ROM1 solution	offline/online	cheap baseline
NN training	offline	amortize nonlinear correction
Surrogate likelihood	online	fast target evaluations
Direct SEP-HMC	benchmark	small-model validation

The win is not that SEP becomes cheap.
The win is that SEP is no longer called at every HMC step.

What Would a Full SEP–NN Run Take?

On this workstation, budget about one week for the production run.

Component	Measured anchor	Production budget
Base SEP data	1,664 corrected unit-shock samples in 2.8 hours	2–4 days for 125 parameter blocks at longer sample and horizon
ZLB SEP data	1,536 binding samples in 6.3 hours	2–5 days to obtain about 5,000 accepted binding points
Combine and train	3,200 points trained quickly; NN lowers ROM1 residual RMSE by 87–94 percent	Less than 2 hours unless the dataset is much larger
Surrogate HMC	100 warmup + 100 draws in 25 minutes at tree depth 5	3–8 hours per 1,500-iteration chain

If base and ZLB data jobs run in parallel, the end-to-end wall-clock estimate is **5–8 days**, including training and matched HMC checks.

The upper tail is solver risk: difficult SEP points, retries, and initialization matter more than neural-network training time.

What Is Validated Today

What is proven today, and what is not.

Validated components

- Corrected unit-shock HLT MVP: 3,200 direct SEP training points.
- ROM1-residual NN cuts observable residual RMSE by 87–94 percent.
- Small-model ELB/OBC matched HMC diagnostics.
- Reduced HLT bridge grid: direct SEP, ROM1, and surrogate surfaces.
- Full-NN and linear+gate MVP HMC smokes run without divergences.

Still open

- Full 18-parameter SEP–NN posterior run.
- Larger prior/posterior support for the corrected unit-shock dataset.
- Robust SEP initialization for the hardest binding points.
- Formal accounting for surrogate error in posterior uncertainty.

Practical Lessons

The practical lesson is discipline, not automation.

1. **Train the correction, not the whole model.** ROM1-residual learning is easier to audit than a direct map.
2. **Report solver coverage.** Failed SEP points are information, not bookkeeping noise.
3. **Validate where the posterior lives.** Average approximation accuracy is not enough.
4. **Keep the empirical claim conditional.** Medium-scale results are only as strong as the validated support.

Closing Takeaways

The method is promising because it is auditable step by step.

1. The computational obstacle is clear: global nonlinear solves are too expensive for the inner likelihood loop.
2. The proposed solution is simple: use SEP offline, learn the ROM1 residual, and put a differentiable surrogate inside HMC.
3. The small nonlinear validation supports the pipeline.
4. The Smets–Wouters extension shows where the hard work remains: solver stability, surrogate error, and posterior-region coverage.

The remaining work is clear: more robust SEP initialization, tighter posterior-region audits, and broader direct benchmarks.

Thank you

Questions and comments welcome.

Backup: Target and Surrogate Error

$$e_\phi(s,\varepsilon,\theta) = g_\theta^{\mathrm{SEP}}(s,\varepsilon) - \left[g_\theta^{\mathrm{ROM1}}(s,\varepsilon) + r_\phi(s,\varepsilon,\theta)\right].$$

Validation quantity

$$\mathrm{RMSE}_{\mathrm{val}} = \left(\frac{1}{N}\sum_{i=1}^{N} \|e_\phi(s_i,\varepsilon_i,\theta_i)\|^2\right)^{1/2}.$$

The same held-out inputs are evaluated under direct SEP, ROM1, and the surrogate.

Backup: Matched HMC Validation Design

- Same synthetic data.
- Same prior and parameter transforms.
- Same initial values and random seeds.
- Same NUTS settings: warmup, draws, target acceptance, tree depth.
- Compare posterior means, MCSEs, and 90 percent intervals.

Pass criterion

The surrogate posterior should overlap the direct SEP posterior within Monte Carlo uncertainty, with no systematic directional distortion.

Backup: Why SEP?

- Keeps the original nonlinear equations.
- Handles occasionally binding constraints directly.
- Produces a simulation object that can be compared with perturbation.
- Expensive, but parallel over training points.

Tradeoff

SEP is a good offline teacher, but a poor online likelihood engine.

Backup: Posterior Interpretation

- The small-model validation targets direct likelihood comparisons.
- The medium-scale empirical exercise uses an inversion/profiled criterion.
- Surrogate HMC results should be read as criterion-based posterior diagnostics until a full direct-SEP HMC benchmark is feasible.

The method is promising because the validation pieces work, not because every medium-scale empirical caveat is already solved.